

Chapter 3

Summarizing Data

The purpose of this section is the description of data using numerical and graphical methods. Numerical methods include measures of central tendency and dispersion. Graphical methods presented are histograms, box plots, and empirical cumulative distribution functions. The last part presents the concepts of correlation and covariance. The next chat also introduces plotting and graphical tools in R for visualizing data, followed by an overview of basic statistical concepts and methods that form the foundation for applied data analysis.

3.1 Measures of Central Tendency

The three main measures of central tendency are mean, median, and mode. The mean reports the average value of a data set. Sometimes, it is also called arithmetic mean or average and can be expressed as follows:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^N x_i$$

The mean does not always explain data very well and there are some other measures to explore. Among them are the median and mode. The median divides the data set into two equal parts, such that half of all the values are greater than the median and half are smaller than the median. The middle term of the data set is the $(n + 1)/2$ -th term. The median has the advantage that a very high or very low values do not influence the median. For example, the average taxable income in Switzerland is \$56,805. The town Vaux-sur-Morges has an average taxable income of \$670,046. It turns out that one very wealthy person (Andre Hoffmann from Roche Pharmaceutical) drives up the average in the village with 178 inhabitants. The mode is the value that occurs the most frequently. Consider the table below on the income distribution of ten citizens

Table 3.1: Income distribution among citizens in three states.

A	B	C
10	2	2
10	3	2
10	7	2
10	8	2
10	9	2
10	10	2
10	10	2
10	14	2
10	17	2
10	20	82

in three states. Note that the mean income in all three states is 10. Although, there is considerable variation among the citizens. The median is 10, 9.5, and 2 for state A, B, and C, respectively. The mode is 10, 10, and 2 for state A, B, and C, respectively.

3.2 Measures of Dispersion

The easiest measure of dispersion is called the range which is the largest value minus the smallest value in the data set. A measure used far more often is the variance. Think of the variance as a measure describing how far the data is spread around the mean. Recall from Chapter 1 that there we distinguish between a population and a sample. The population is characterized by unknown parameters and we use statistics based on sample to say something about the unknown population parameters. The distinction between population and sample is important regarding the variance. The population variance is calculated as follows:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

The sample variance is calculated as follows:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

The population variance is used to calculate the variance of the entire population. For example, if a professor is interested in calculating the variance associated with the final exam, they would use population variance equation because the class is not a sample of a larger population. The second equation dividing by

$N - 1$ is used if we have a sample and want to estimate (!) the variance associated with the population. Consider the figure below which illustrates the difference. A randomly generated data set contains 100,000 observations with $\mu = 50$ and $\sigma = 20$. An simulation procedure takes 1000 sample with sample sizes varying from 2 to 50 (horizontal axis). For each of the sample, the population variance is estimated by dividing by either N or $N - 1$. The estimates dividing by $N-1$ are closer to the population standard deviation than for the samples divided by N .

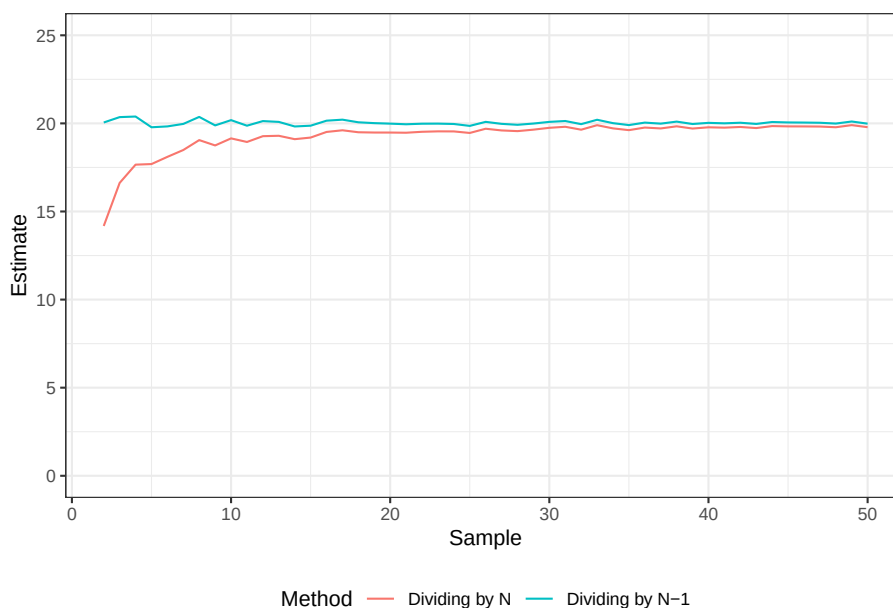


Figure 3.1: Difference between dividing by N and $N-1$ to estimate the variance.

The coefficient of variation standardizes the standard deviation σ by the mean, i.e., $CV = \sigma/\mu$. Because the magnitude of the standard deviation depends on the mean, it is sometimes necessary to calculate the coefficient of variation to make two or more standard deviations comparable. For example, suppose you are comparing residential home values in California and Indiana. You calculate the mean and standard deviation for California as \$2,000,000 and \$400,000, respectively. The mean and standard deviation for Indiana are \$125,000 and \$50,000, respectively. Calculating the coefficient of variation (CV) for both states leads to $CV_{CA} = 0.2$ and $CV_{IN} = 0.4$. Hence, there is much more variation in the home values for Indiana than there is in California.

3.3 Histograms

In almost all cases, a visual inspection of the data is appropriate. Although numerical statistics such as mean and/or standard deviation exist, many fail to easily identify patterns or anomalies. Histograms are probably the most basic method to graphically summarize data and are also good approximations of the probability distributions. Suppose you have a data set $x_1, x_2, \dots, x_n$. You divide the range of values into bins. For example, if a data set ranges from 75 to 155, then a possible bin size could be 10, i.e., 75, 85, $\dots$, 155. The height of the bar for each bin is determined by the number of observations in the bin range. Those bins usually have the same size but it is not necessary. Consider the eruption and waiting times of Old Faithful geyser in Yellowstone National Park. Panel (a) and (b) represent the histograms of eruption and waiting time, respectively.

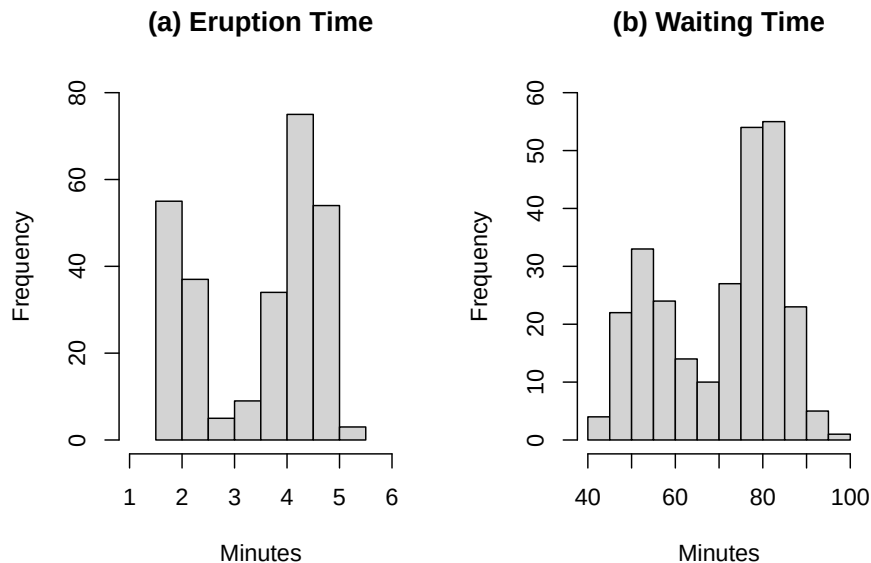


Figure 3.2: Histogram of eruption and waiting time of Old Faithful geyser

To draw a histogram of the data as shown in the above figure, use the following command:

- `hist(faithful$eruptions)`

Note that you can control the appearance of the histogram by using options:

- `hist(faithful$eruptions, main="Eruption", xlab="Minutes", xlim=c(1,6), ylim=c(0,80))`

3.4 Empirical Cumulative Distribution Function

The empirical cumulative distribution function can be written as

$$F_n(x) = \frac{\text{number of elements} \leq x}{n}$$

Given the previously mentioned Old Faithful data, the empirical cumulative distribution function is shown below. An important measure of empirical cumulative distribution functions are quantiles. Quantiles are the values that cut $x\%$ of the distributional area. Below the quantile is a certain share of all values in a set of ordered observations. If the distribution is divided in n equal shares, there are $n-1$ quantiles. Commonly used quantiles are quartiles, quintiles, deciles, and percentiles. Quartiles divide the sample into 4 equal shares, i.e., 3 quartiles: 25%-, 50%-, 75%- quartile. Quintiles divide the sample into 5 equal shares, deciles into 10 equal shares, and percentiles divide the sample into 100 equal shares. Closely related to quartiles is the *interquartile range (IQR)*.

```
faithful = data.frame(faithful)
par(mfrow=c(1,2))
plot(ecdf(faithful$eruptions),main="Eruption Time",xlab="Minutes",xlim=c(1,6),lty=1)
plot(ecdf(faithful$waiting),main="Waiting Time",xlab="Minutes",xlim=c(40,100),lty=1)
```

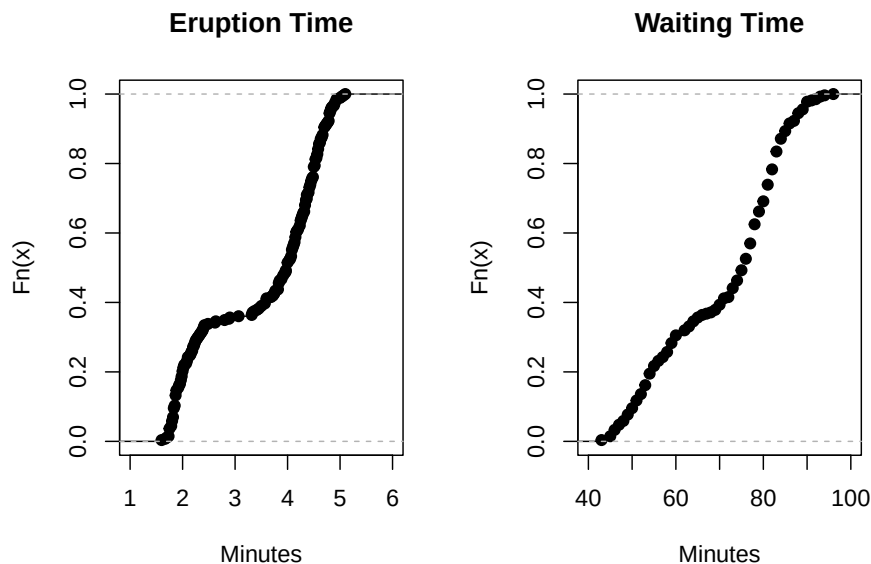


Figure 3.3: Empirical cumulative distribution functions of eruption and waiting time of Old Faithful geyser

3.5 Boxplots

Outliers are values that extremely deviate from the mean, i.e., extremely high or low values. Those values limit the explanatory power of the mean. Reasons for outliers can be error in measurement, error of judgement, miscalculation, typo in data matrix, real extraordinary values. To treat outliers, we must know whether they are true values or a mistake. Outliers can be neglected under certain conditions, but it is important to bear them in mind when interpreting results!

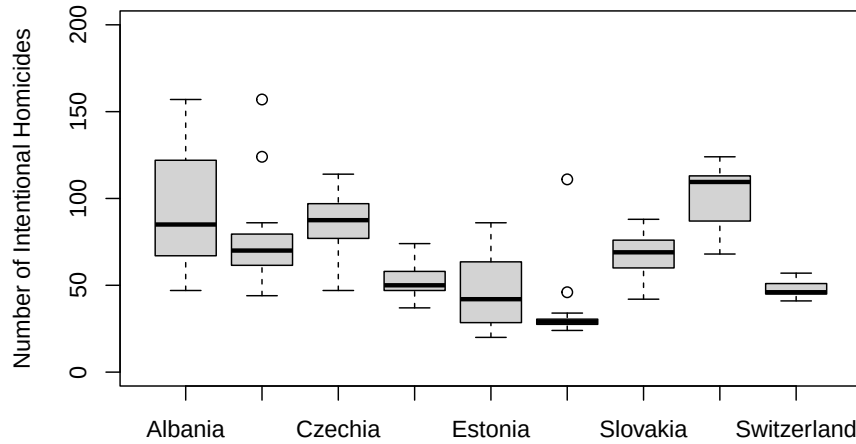


Figure 3.4: 2008-2022 Intentional homicides in select European countries, i.e., Albania (AL), Austria (AT), Switzerland (CH), Czech Republic (CZ), Denmark (DK), Estonia (EE), Norway (NO), Sweden (SE), and Slovakia (SK) Source: Eurostat

3.6 Covariance and Correlation Coefficient

The previous sections focused on one random variable at a time. Very often, we have more two or more random variable and we are interested in their relationship. We will analyze the relationship between variables from a causal standpoint in the section on regression analysis. In this chapter, we focus on two variables and how they behave jointly. For now, we will not make any statements about causality. The important part here: Correlation is not causation! As for the variance, there are two definitions/equations to calculate the covariance between

two variables:

$$\text{Cov}(x, y) = E[(x - E(x)) \cdot (y - E(y))] = E(x \cdot y) - E(x)E(y)$$

If the sign of the covariance is positive, then x and y tend to move in the same direction, i.e., if one variable increases, the other variable increases as well. If the sign of the covariance is negative, then x and y tend to move in opposite directions, i.e., if one variable decreases, the other increases. If X and Y are independent, then $\text{Cov}(X, Y) = 0$. The covariance has several properties:

- Property 1: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X, Y)$
- Property 2 (Transformation of the covariance): $\text{Cov}(r \cdot X + s, t \cdot Y + u) = r \cdot t \cdot \text{Cov}(X, Y)$

The most important aspect about correlation (and statistics in general): Correlation does not mean causation. Causation requires a strong theoretical believe that one variable is the cause of another variable, e.g., influence of education on income. The correlation coefficient (sometimes called Pearson's r) is defined as

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

The correlation coefficient varies between -1 and 1 . The Sign provides the direction of the relationship between two variables and the Value provides the magnitude of the relationship. Note that the correlation coefficient has no dimensions!

A second example uses the data `mh2` and the resulting scatter plot is shown below.

The function `summary()` gives you mean, median, and quartiles for the eruption and waiting times. The functions `var()` and `cor()` calculate the variance, covariance, and correlation coefficient associated with the data sets. We will see in subsequent chapters how to interpret the covariance and correlation coefficients.

```
faithful = data.frame(faithful)
summary(faithful)
```

```
##      eruptions      waiting
## Min.   :1.600   Min.   :43.0
## 1st Qu.:2.163   1st Qu.:58.0
## Median :4.000   Median :76.0
## Mean   :3.488   Mean   :70.9
## 3rd Qu.:4.454   3rd Qu.:82.0
## Max.   :5.100   Max.   :96.0
```

```
var(faithful)
```

```
##      eruptions      waiting
```

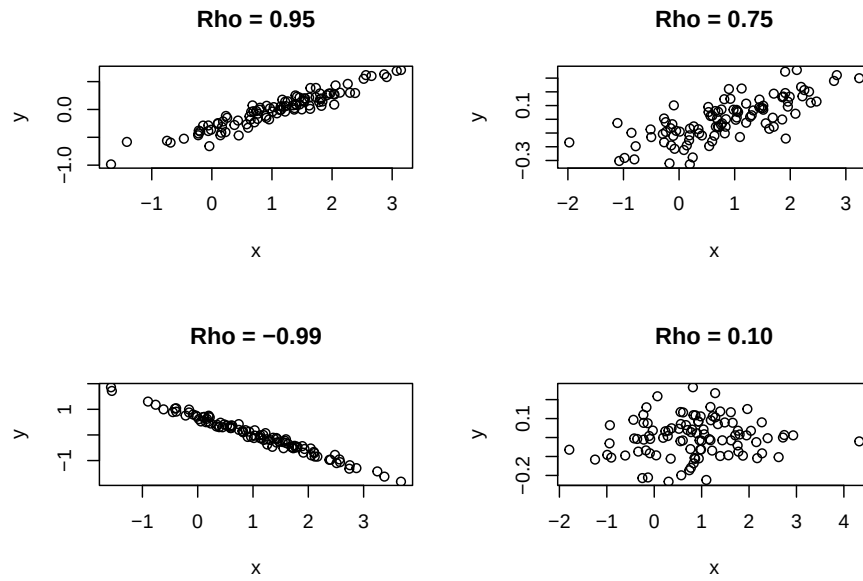


Figure 3.5: Examples of various correlation coefficients

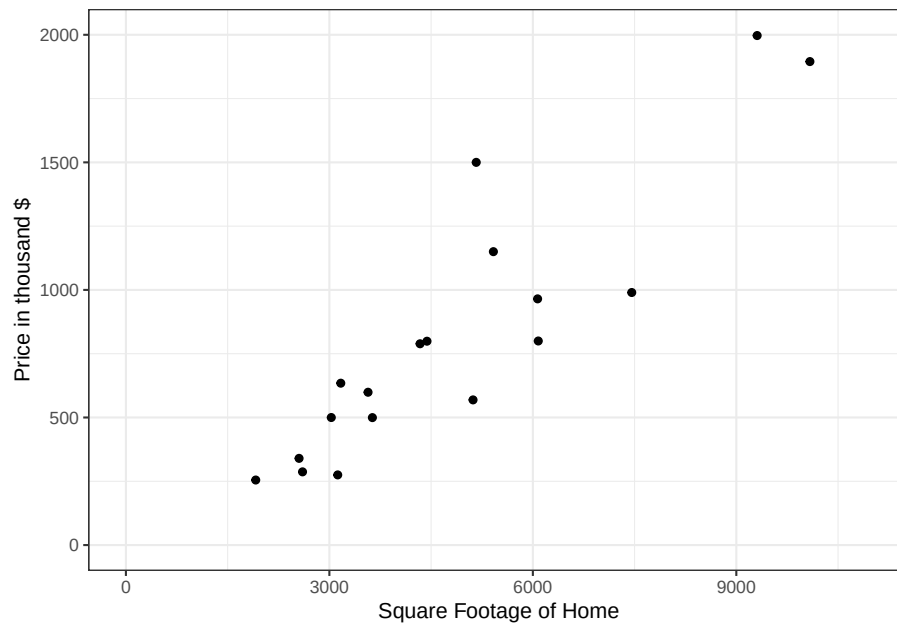


Figure 3.6: Correlation between the square footage of a home and the price of the home in the Meridian Hills neighborhood in Indianapolis.

```
## eruptions  1.302728  13.97781
## waiting   13.977808 184.82331
cor(faithful$eruptions,faithful$waiting)
```

```
## [1] 0.9008112
```

Note that sometimes, we deal with qualitative data, i.e., data that is not expressed as a number but as an expression. Example are gender (male/female), owning a car (yes/no), modes of commute (car, bike, train, bus, etc.). Consider the data in `gssgun`. One way to count the responses for firearm ownership is to use the following command:

- `table(gss$owngun)`

Suppose you are only interested in people who answer *yes* or *no* for the questions concerning arrests and firearms ownership. You will have to subset the data.

```
df = subset(gss,year==2022,select=c("owngun","sex"))
CrossTable(df$owngun,df$sex,prop.chisq=FALSE)
```

```
##
##
##      Cell Contents
## |-----|
## |                      N |
## |      N / Row Total |
## |      N / Col Total |
## |      N / Table Total |
## |-----|
##
##
## Total Observations in Table:  2301
##
##
##      | df$sex
## df$owngun |      1 |      2 | Row Total |
## -----|-----|-----|-----|
##      1 |      424 |      331 |      755 |
##      |      0.562 |      0.438 |      0.328 |
##      |      0.395 |      0.270 |      |
##      |      0.184 |      0.144 |      |
## -----|-----|-----|-----|
##      2 |      623 |      879 |      1502 |
##      |      0.415 |      0.585 |      0.653 |
##      |      0.580 |      0.716 |      |
##      |      0.271 |      0.382 |      |
## -----|-----|-----|-----|
```

```
##           3 |           27 |           17 |           44 |
##           |           0.614 |           0.386 |           0.019 |
##           |           0.025 |           0.014 |           |
##           |           0.012 |           0.007 |           |
## -----|-----|-----|-----|
## Column Total |           1074 |           1227 |           2301 |
##           |           0.467 |           0.533 |           |
## -----|-----|-----|-----|
##
##
```

3.7 Exercises

1. **Natural Gas Usage and Temperature** (**): Use the data `indyheating` for this exercise. Construct a scatter plot with temperature on the horizontal axis and natural gas usage on the vertical axis. What do you observe? How would you expect the graph to look like if electricity consumption was on the vertical axis?
2. **Exempt Organizations** (***) : Consider the data in `exemptorgs`. You are going to calculate the average revenue across the so-called NTEE codes. Proceed as follows:
 - a. Subset the data such that you are only left with the columns *revenue* and *ntee*.
 - b. Use the function `na.omit()` on the data set you created in (a). What is the function used for?
 - c. Use the function `aggregate()` and calculate the mean assets, revenue, and income by NTEE code. The table created in your answer should contain the NTEE codes in plain English, that is, not the alphabetical codes.
3. **GSS Two-Way Table** (**): In the data set `gss`, pick two of the following variables: *fulltime*, *government*, *married*, *vote*, *gun*, *deathpenalty*. Construct a cross-table similar to the one about gender and guns in the section *Introduction to R*. using the function `CrossTable()`. Ignore the Chi-square statistic but explain if you see any pattern that is of interest.
4. **Airport Delays Boxplot** (***) : Pick an airport (not IND) and year of your choice in the data set `airlines`. You should be using the function `subset()` to pick year and airport. Next, add a column called *delay* which is the share of delays from all the arriving flights. Next, construct a boxplot with all the airlines on the x-axis, i.e., one boxplot, and the variable *delay* on the vertical axis. Interpret the boxplot. Are there airlines which are particularly on time or always late?
5. **Housing Price Index** (***) : Pick one of 49 U.S. States (not Indi-

ana). Go to the [FRED webpage](#) and download two data series: (1) All-Transactions House Price Index for the United States (USSTHPI) and (2) All-Transactions House Price Index for the state of your choice. Answer the following questions.

- What do the two data series describe?
- Plot the two data series over time. Your result should look similar to Panel (b) in Figure ??.
- How do the housing prices evolve in your state compared to the United States. Do homes in your state get more expensive than the general trend? Less? How has the housing market evolved during and after the 2008 recession?

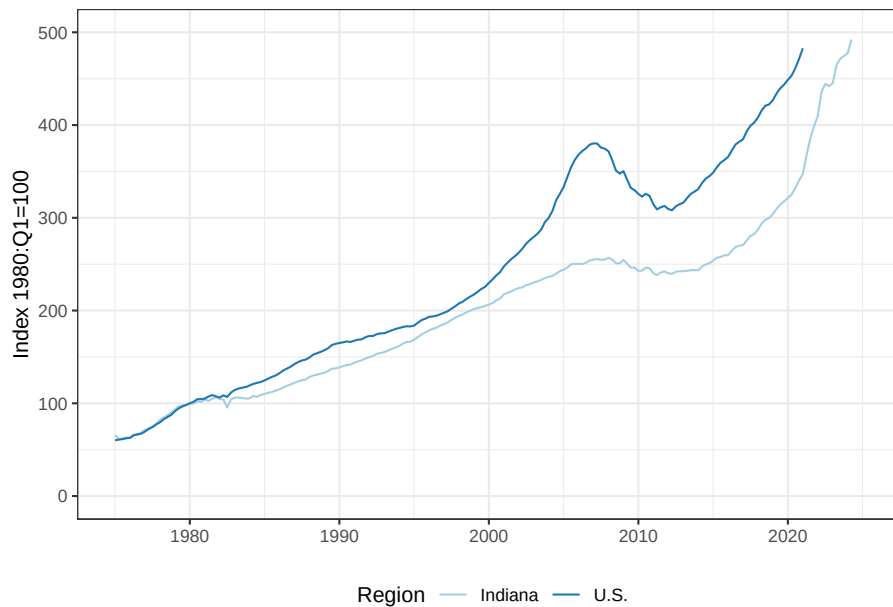
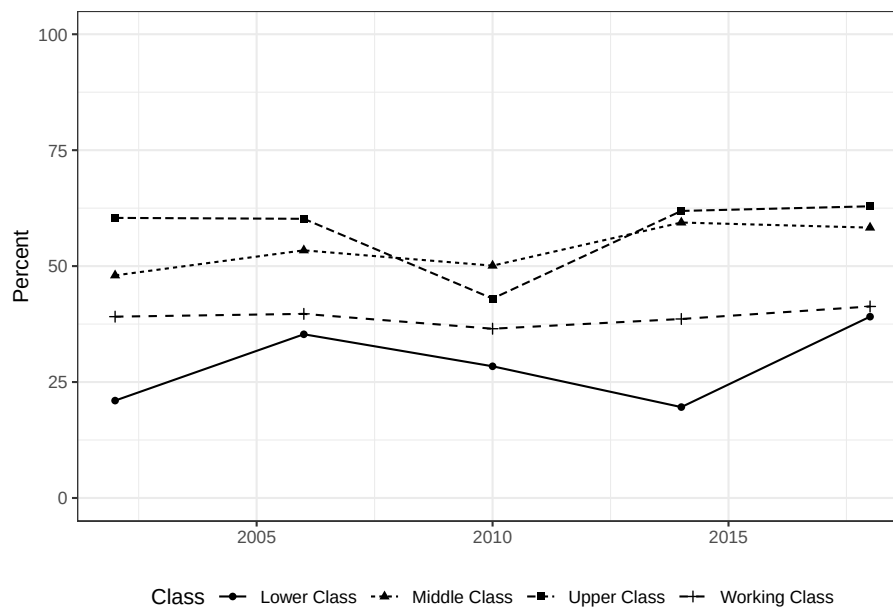


Figure 3.7: Evolution of the All Transaction House Price Index for the U.S. and Indiana (Source: FRED St. Louis)

6. **BMW Boxplot** (**): Consider the used car data set `bmw` for this exercise which contains the prices and miles of a particular BMW model in the Indianapolis area. The column `allwheeldrive` indicates whether the car has all wheel drive (1) or not (0). You must use the R/RStudio commands and not just look at the data. Answer the following questions:
 - a. Calculate the following statistics for price and miles: Minimum, maximum, median, and mean.
 - b. Calculate the same statistics as in the previous part but separate the data into two groups: (1) with all-wheel drive and (2) without

- all-wheel drive.
- c. Use the data on prices only. Create a box-and-whisker plot grouped by all-wheel drive. This must be one graph.
7. **State Capitals** (**): The data `apartments` contains data about the monthly rent and size of furnished apartments in Berlin and Potsdam, which are not only adjacent cities but are also capitals of two states in Germany. Add a column that shows the rent per square meter. Then, construct a box plot of the rent per square meter on the y -axis and the location on the x -axis. Based on the boxplot, how does the rent per square meter differ between the two cities. Any other observation, which you are able to make?
8. **EPA Fuel Economy** (**): Use the 2018 data contained in `vehicles` for this problem.
- a. Generate a scatter plot of the variables `displ` and `comb08U`. What can you say about the shape of the scatter plot and the relationship between engine displacement and fuel economy.
 - b. Transform both variables into their natural logarithm and plot the scatter plot. What changes?
 - c. Create a table summarizing the average fuel economy by vehicle class (`vclass`) for the following four manufacturers: (1) Ford, (2) Chevrolet, (3) Toyota and (4) Honda. You will have to use the function `aggregate()` for this.
9. **GSS Trends** (**): The General Social Survey is an annual survey tracking societal trends in the United States. For a more detailed description, you can go to [here](#). In this exercise, I want you pick a particular issue and one breakdown (e.g., age, sex, political affiliation) and plot a graph with R/RStudio on how attitudes about the issue have evolved over time. I suggest, you go to [trends](#) and pick a topic you are interested in. You will see that the website already provides you with a trend graph but I want you to use R/RStudio for this exercise and re-create the graph (i.e., you get zero points if you simply copy the graph provided on the website). For example, suppose you are interested in the variable `rincblls` which asks the respondent “Do you feel that the income from your job alone is enough to meet your family’s usual monthly expenses and bills?” One of many possibility to plot this with R/RStudio would look like Panel (b) in Figure ?? . Make sure to label the x,y -axis correctly as well as provide a legend for the graph.



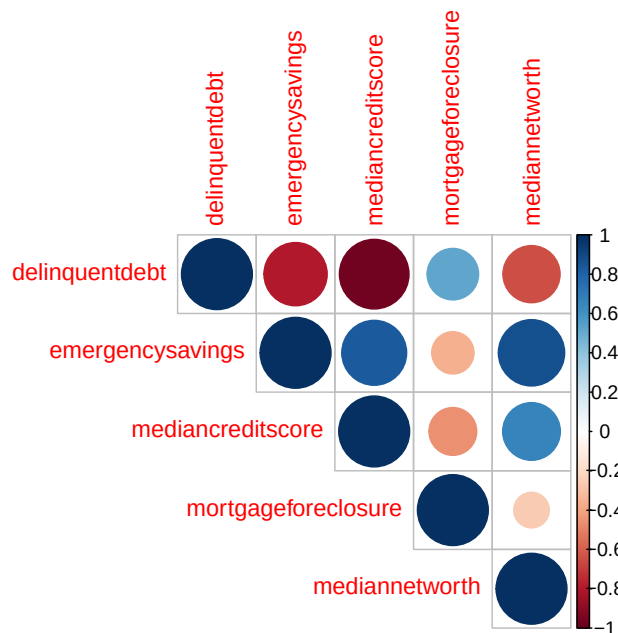
10. **WDI Boxplot** (**): The World Bank data `wdi` contains development indicators and also information how the World Bank classifies those countries by region and income. Focus on the years 1975, 1985, 1995, 2005, and 2015 for this question. Use the command `subset()` to extract those years. Next, you have to install and load the package `ggplot2`. The package `ggplot2` allows you to do some great data visualization. Execute the commands below. The result is a boxplot by region and by year. Based on the resulting plot, explain differences between regions in terms of life expectancy and also in terms of evolution over time.

```
wdiofinterest = subset(wdi, year %in% c(1975, 1985, 1995, 2005, 2015))
ggplot(wdiofinterest, aes(x=region, y=lifexp, fill=as.character(year))) +
  geom_boxplot(position=position_dodge(1))
```

11. **Faithful** (**): The data set `faithful` which is included with R contains data about the eruption and waiting times in minutes between eruptions from Old Faithful geyser in Yellowstone National Park. Use a scatter plot to visualize the relationship between eruption and waiting time. That is, generate graph with *eruption* on the vertical axis and *waiting* on the horizontal axis. What do you observe? What is the correlation coefficient? Is there anything odd in the resulting scatter plot? Next, use R to construct an empirical cumulative distribution function.
12. **Financial Health** (**): Consider the data in `statefinhealth`. Execute the following commands requiring the package `corrplot` and include the resulting figure in the homework. What does the plot display? Explain the

relationship between the components. Are some of the results surprising?

```
dffin = statefinhealth[,c(2:6)]
dffin = cor(dffin)
corrplot(dffin,type="upper")
```



13. **Earnings (***)**: Consider the data in `earnings`, which contains the real earnings by education level, sex, and race/ethnicity. Generate a line plot using the package `ggplot2` with year on the horizontal axis and the earnings (*value*) on the vertical axis. Consider “All Races” but differentiate by education level and sex. That is, differentiate the various education levels with colors and use the `ggplot` component “`+facet_grid()`” to plot three graphs side-by-side, i.e., both sexes, men, women. Repeat the exercise using the variable *growth*.
14. **Power Generation (***)**: Each year, the U.S. Energy Information Administration (EIA) publishes data about all electric generators in the United States. You will find selected variables for all observations of that data in `eia`. Use the function “`unique()`” from R to determine the technologies used in the EIA report. You will see that there are multiple technologies using wind and solar (e.g., offshore wind, onshore wind). The goal of this exercise is plotting the share of wind and solar from the total generation capacity in the state. For this exercise, you will have to use multiple functions, e.g., `aggregate()`, `merge()`, to arrive at the final result. Plot the resulting share using the function “`ggplot()`” from R. California is usually considered the leader in renewable power generation. Is that still

true when considering the share of installed power?